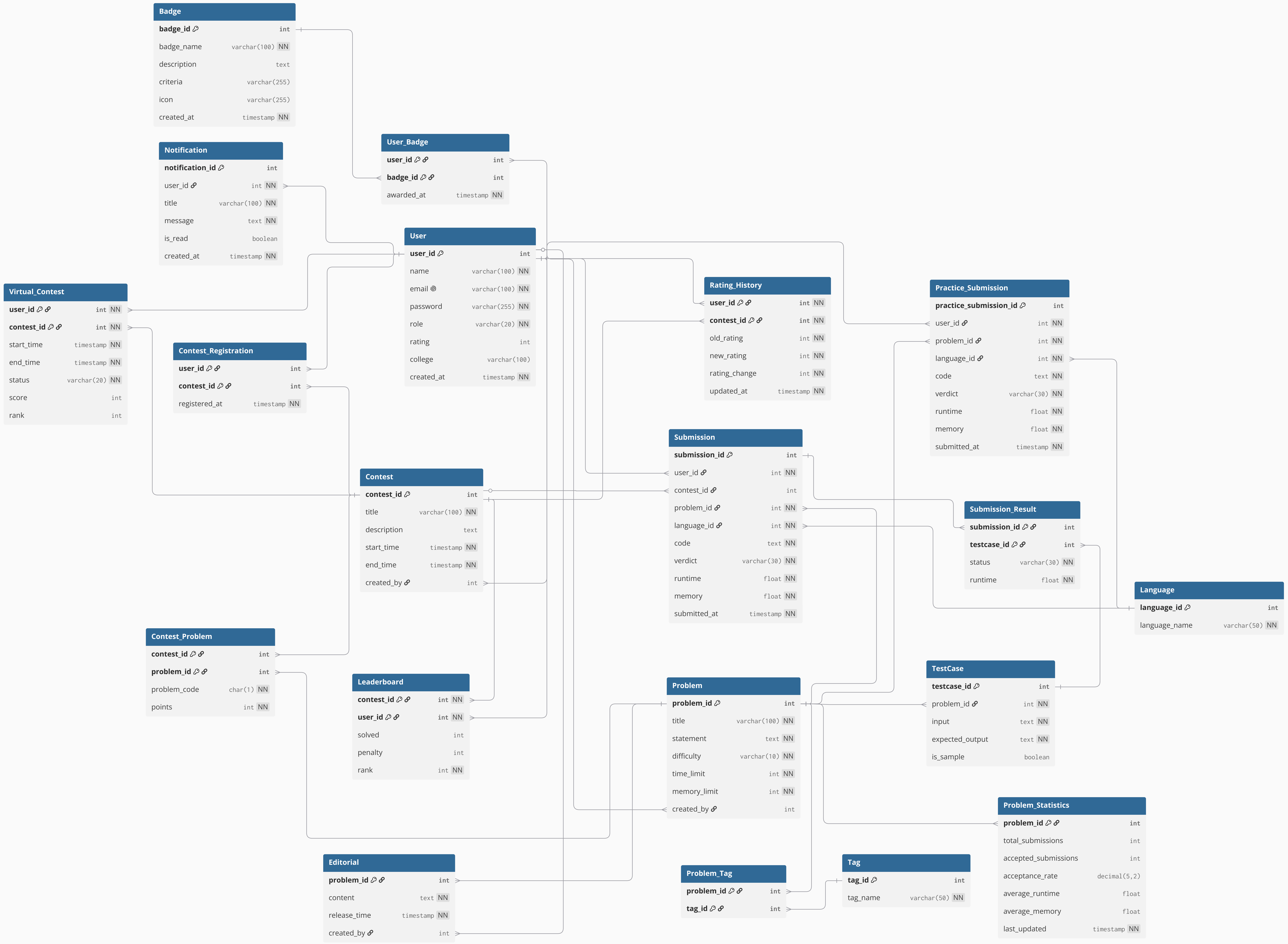

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Proof that Relations are in BCNF

A relation R is in Boyce-Codd Normal Form (BCNF) when the determinant of every functional dependency (FD) that holds on R is a super key of R. In other words, for every FD A → B that holds on relation R, A must be a super key.

1. User Table

- FDs: `user_id → name, email, password, role, rating, college, created_at`
    - `user_id` is the primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

2. Language Table

- FDs: `language_id → language_name`
    - `language_id` is the primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

3. Tag Table

- FDs: `tag_id → tag_name`
    - `tag_id` is the primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

4. Badge Table

- FDs: `badge_id → badge_name, description, criteria, icon, created_at`
    - `badge_id` is the primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

5. Contest Table

- FDs: `contest_id → title, description, start_time, end_time, created_by`
    - `contest_id` is the primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

6. Problem Table

- FDs: `problem_id → title, statement, difficulty, time_limit, memory_limit, created_by`
    - `problem_id` is the primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

7. Contest_Problem Table

- FDs: `(contest_id, problem_id) → problem_code, points`
    - `(contest_id, problem_id)` is the composite primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

8. Contest_Registration Table

- FDs: `(contest_id, user_id) → registered_at`
    - `(contest_id, user_id)` is the composite primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

9. TestCase Table

- FDs: `testcase_id → problem_id, input, expected_output, is_sample`
    - `testcase_id` is the primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

10. Submission Table

- FDs: `submission_id → user_id, contest_id, problem_id, language_id, code, verdict, runtime, memory, submitted_at`
    - `submission_id` is the primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

11. Submission_Result Table

- FDs: `(submission_id, testcase_id) → status, runtime`
    - `(submission_id, testcase_id)` is the composite primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.
    -

12. Leaderboard Table

- FDs: `(contest_id, user_id) → solved, penalty, rank`
  - `(contest_id, user_id)` is the composite primary key, and all non-key attributes depend on it.
  - Conclusion: This relation is in BCNF.

13. Problem_Tag Table

- FDs: `(problem_id, tag_id) → ∅`
  - `(problem_id, tag_id)` is the composite primary key. This table contains only key attributes and has no non-key attributes.
  - Conclusion: This relation is in BCNF.

14. Editorial Table

- FDs: `problem_id → content, release_time, created_by`
  - `problem_id` is the primary key, and all non-key attributes depend on it.
  - Conclusion: This relation is in BCNF.

15. Virtual_Contest Table

- FDs: `(user_id, contest_id) → start_time, end_time, status, score, rank`
  - `(user_id, contest_id)` is the composite primary key, and all non-key attributes depend on it.
  - Conclusion: This relation is in BCNF.

16. Rating_History Table

- FDs: `(user_id, contest_id) → old_rating, new_rating, rating_change, updated_at`
  - `(user_id, contest_id)` is the composite primary key, and all non-key attributes depend on it.
  - Conclusion: This relation is in BCNF.

17. Practice_Submission Table

- FDs: `practice_submission_id → user_id, problem_id, language_id, code, verdict, runtime, memory, submitted_at`
  - `practice_submission_id` is the primary key, and all non-key attributes depend on it.

    - Conclusion: This relation is in BCNF.

18. Notification Table

- FDs: `notification_id → user_id, title, message, is_read, created_at`
    - `notification_id` is the primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

19. User_Badge Table

- FDs: `(user_id, badge_id) → awarded_at`
    - `(user_id, badge_id)` is the composite primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.

20. Problem_Statistics Table

- FDs: `problem_id → total_submissions, accepted_submissions, acceptance_rate, average_runtime, average_memory, last_updated`
    - `problem_id` is the primary key, and all non-key attributes depend on it.
    - Conclusion: This relation is in BCNF.